STOREPULSE AI - LIVE DEMO SCRIPT

TalentAP Mega Hackathon 2026 - Problem Statement R5

1. PROJECT OVERVIEW

StorePulse AI turns unstructured retail feedback into operational action.
The platform processes multilingual store feedback and delivers:
• Issue classification (multilingual: English, Hinglish, Telugu, Tamil, Marathi)
• Policy-first safety escalation (allergens, contamination, hygiene, misconduct)
• Department routing (Store Operations, Supply Chain, Merchandising, Digital, HR)
• Emerging incident detection (early-warning store-level spike detection)
• Weekly regional action briefs (PDF/CSV reports with evidence quotes)
• Cross-channel duplicate detection
• Evaluation & model insights

2. TECHNOLOGY STACK

Frontend/Framework: Streamlit (app.py) + Next.js 16 (React 19)
Data Processing: Pandas, NumPy
NLP/ML: Sentence-Transformers (optional), Scikit-learn
Rule-based engine: Transparent keyword + policy patterns
Visualization: Plotly
Database: SQLite (demo) + PostgreSQL (Drizzle ORM)
Reporting: ReportLab (PDF), Pandas CSV

3. DEMONSTRATION OF KEY FEATURES

3.1 AI FEEDBACK ANALYZER
Classifies multilingual feedback in real-time
Demonstrating with sample feedback:
1. [REFRIGERATION FAILURE] Severity=HIGH Actionable=True Dept=Merchandising Method=rule_based
2. [ALLERGEN LABEL ERROR] Severity=HIGH Actionable=True Dept=Merchandising Method=rule_based
3. [GENERAL FEEDBACK] Severity=MEDIUM Actionable=True Dept=Store Operations Method=rule_based
4. [STAFF MISCONDUCT] Severity=HIGH Actionable=True Dept=HR Method=rule_based
3.2 POLICY-FIRST SAFETY ESCALATION
Safety categories always escalate to HIGH severity
'Milk packet has nuts but no allergen label...' → Severity=HIGH Escalates=True Dept=Merchandising
'Staff harassment at store entrance...' → Severity=HIGH Escalates=True Dept=HR
'Generic positive feedback from customer...' → Severity=MEDIUM Escalates=False Dept=Store Operations
3.3 EMERGING INCIDENT DETECTION
Early-warning system based on normalized complaint rates
272 alerts detected from 3692 records
• Store ST-031 | App Payment Failure Growth=4.00x Count=2
• Store ST-022 | Stock Shortage Growth=3.50x Count=2
• Store ST-036 | Equipment Failure Growth=2.80x Count=2
3.4 AUTOMATIC DEPARTMENT ROUTING

'refrigeration failure' → Department: Store Operations Action: Inspect refrigeration equipment and isolate affected products.
'allergen' → Department: Store Operations Action: Review the issue with the owning team.
'payment failure' → Department: Store Operations Action: Review the issue with the owning team.
'staff misconduct' → Department: HR Action: Escalate to HR and conduct an incident review.
3.5 CROSS-CHANNEL DUPLICATE DETECTION
319 potential duplicate groups identified
Potential duplicate incident reported across up to 3 channels.
3.6 WEEKLY REGIONAL ACTION BRIEFS
Generates downloadable PDF/CSV reports per region
Includes: regional overview, top 5 priority issues, action suggestions,

4. INTERACTIVE UI (STREAMLIT AT LOCALHOST:8501)

The live app features a sidebar with:
• Navigation between 7 pages: Dashboard, AI Analyzer, Emerging Alerts, Weekly Reports, Store Analytics, Evaluation, Settings/About
• Filter controls: Region, Store, Channel, Severity, Issue Category, Date Range
• KPI metrics: Total feedback, High-severity issues, Emerging incidents, Stores monitored, Actionable %, Departments requiring attention
• Demo mode: 4 judge-ready scenarios that run through live application logic
• Settings: Toggle embeddings, upload official datasets, replace taxonomy
• Download buttons: PDF and CSV reports per region
• Page-specific: confusion matrices, model insights, trend charts

5. DEMO SUMMARY

StorePulse AI demonstrates how transparent rule-based AI can solve real retail operations problems without paid APIs. All classification decisions are explainable: every alert shows which rules matched, which policies triggered, and which department owns the issue.
Key hackathon values:
• Multilingual support (English, Hinglish, transliterated Indian languages)
• Policy-first severity (safety categories → HIGH automatically)
• Emerging detection with robust baselines (median/MAD)
• Zero-history store safety (handles new stores with low volume)
• Full explainability (no fabricated confidence scores)
• Works offline with synthetic demo data clearly marked

6. RUNNING THE APPLICATION

Streamlit app: http://localhost:8501
Demo script: python live_demo_script.py
Synthetic data: 3,692 feedback records with seeded incidents
Thresholds: minimum_support=2, growth_threshold=2.5, spike_threshold=3.0